

DSE/GE4b: Introduction to Web Programming

Comprehensive Study Notes

Units 1, 2 & 3 | NEP UGCF 2022 | Semester VI/IV

Based on: Responsive Web Design with HTML5 & CSS (9th Ed.) – Jessica Minnick

UNIT 1: INTRODUCTION

Introduction • Exploring the Internet • Planning a Website

1.1 The Internet

The Internet is a worldwide collection of computers linked together using modems, phone lines, television cables, satellite links, fiber-optic connections, and radio waves. It was developed in the 1960s by the U.S. Department of Defense Advanced Research Projects Agency (ARPA). The original network, ARPANET, sent its first message in 1969 with only four nodes.

Key Terms – Internet Infrastructure

Node: Any device (computer, tablet, smartphone) connected to a network.

Network: A collection of two or more computers linked to share resources and information.

Internet Backbone: A collection of high-speed data lines linking major computer systems worldwide.

ISP (Internet Service Provider): A company with a permanent connection to the Internet backbone that provides access to businesses and individuals.

Internet of Things (IoT): The ever-growing number of everyday devices (TVs, appliances, etc.) connecting to a network.

1.2 World Wide Web

The Internet and the World Wide Web are NOT the same thing. The Internet is the physical infrastructure (networks of computers). The World Wide Web (web) is a SERVICE running on top of the Internet that provides access to information stored on web servers.

Key Web Terms

Web Server: A high-capacity, high-performance computer that stores and delivers webpages.

Webpage: A file that is part of the web, capable of displaying text, graphics, audio, and video.

Website: A related collection of webpages created by a person, company, or organization.

Home Page: The main/first page of a website; provides information about the site's purpose and links to other pages.

Hyperlink (Link): An element that connects one webpage to another on the same or different server. Text, graphics, or animations can serve as links.

1.3 Internet Protocols

A protocol is a set of rules that defines how computers communicate with each other. A client workstation uses protocols to request connections to servers.

Protocol	Full Name	Purpose
HTTP	Hypertext Transfer Protocol	Fundamental protocol for exchanging webpages on the web. Used every time you click a link.
HTTPS	HTTP Secure	Encrypted version of HTTP for secure data transmission.
FTP	File Transfer Protocol	Used to exchange files between computers over the Internet (not the web). Used by web designers to upload files to web servers.
TCP/IP	Transmission Control Protocol / Internet Protocol	Pair of protocols for efficiently routing data across the Internet. TCP breaks data into packets; IP ensures correct delivery.
IP Address	Internet Protocol Address	Unique numerical address for every device on the Internet (e.g., 192.168.1.5).
DNS	Domain Name System	Associates IP addresses with human-readable domain names (e.g., 204.79.197.200 ↔ bing.com).

1.4 Web Browsers

A web browser is a program that interprets and displays webpages so you can view and interact with them. Popular browsers include: Microsoft Edge, Mozilla Firefox, Google Chrome, Apple Safari, and Opera.

Key browser functions:

- Locate and display websites
- Navigate between webpages using links
- Bookmark/favorite webpages
- Manage security settings
- Interpret HTML and CSS to render pages visually

1.5 HTML – Hypertext Markup Language

HTML is a markup language that uses tags to define the structure and layout of content in a webpage. It is NOT a traditional programming language. HTML has evolved from version 1.0 (1989) to the current HTML 5.

HTML Elements & Tags

An HTML element consists of everything from the start tag to the end tag, including the content between them.

- **Start tag (opening tag):** <tagname>
- **End tag (closing tag):** </tagname>
- **Content:** Text or other elements between the tags
- Void/Empty elements have no content or end tag:
, <hr>,

```
<title> Webpage Example </title>
```

HTML Attributes

Attributes provide additional information/characteristics about an element. They are placed inside the start tag as name="value" pairs.

```

```

Basic HTML Document Structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Page Title</title>
  </head>
  <body>
    <!-- Page content goes here -->
  </body>
</html>
```

Tag / Element	Purpose
<!DOCTYPE html>	Declares the document type as HTML5
<html>	Root element of the webpage
<head>	Contains metadata (not displayed on page)
<meta charset="UTF-8">	Sets character encoding
<title>	Sets the title shown in browser tab
<body>	Contains all visible page content
<h1> to <h6>	Headings (h1 = largest, h6 = smallest)
<p>	Defines a paragraph
 	Line break (empty element)
<hr>	Thematic break / horizontal rule (empty element)

1.6 Types of Websites

- Personal – blogs, portfolios, family sites
- Organizational / Non-profit – charities, clubs
- Commercial / Business – e-commerce, company sites
- Government – public information and services
- Educational – schools, universities, e-learning
- Informational / Portal – news, search engines, directories
- Entertainment – gaming, streaming, social media

1.7 Web Authoring Tools

Web authoring tools help create and manage webpages. There are two main categories:

Text Editors: Write HTML/CSS code directly (e.g., Notepad++, Visual Studio Code, Sublime Text). Professionals prefer these for full control.

WYSIWYG Editors: What You See Is What You Get. Visual editors like Adobe Dreamweaver that generate code automatically.

1.8 Planning a Website

Good websites require careful planning before any code is written. The planning phase involves identifying the target audience, goals, content, and structure.

Target Audience Analysis

Understanding your target audience helps design decisions about typography, color, layout, and content. Consider:

- Age group and demographics
- Technical comfort level
- Devices used (mobile vs. desktop)
- Goals and needs of the user

Website Design Principles

Accessibility: Design so all users, including those with disabilities, can use the website. Use alt text for images, proper contrast, etc.

Multiplatform Display: Design for mobile, tablet, and desktop screens (responsive web design).

Navigation: Use clear, consistent navigation with tabs, menus, and text links.

Typography: Choose fonts that match the site's purpose and are easy to read.

Color: Use color consistently to reinforce branding and guide the user's eye.

White Space

Active White Space: Intentionally left blank areas used to balance asymmetrical page design.

Passive White Space: Space between content areas that helps users focus on specific content. Proper use improves readability.

Wireframe

A wireframe is a simple visual guide (sketch) that clearly identifies the location of main webpage elements such as the navigation area, logo, content areas, and images. Use lines and boxes to represent elements. Tools: Pencil Project, Mockplus, Wireframe CC, or even pen and paper.

Site Map

A site map is a planning tool that lists or displays all pages on a website and shows how they relate to each other (the structure/hierarchy of the website). Organize content by: product category, step-by-step sequence, topic area, etc.

UNIT 2: HTML

HTML Structure • Images • Div Elements • Links • Lists • Tables • Forms

2.1 HTML Document Structure (Ch. 1–2)

Every HTML document follows a standard structure. HTML 5 uses semantic elements that describe the meaning of content.

Semantic HTML 5 Elements

Element	Description
<header>	Contains introductory content or navigational aids (logo, site name)
<nav>	Contains navigation links
<main>	Main content of the document (unique per page)
<article>	Self-contained content that can stand alone (blog post, news article)
<section>	Thematic grouping of content
<aside>	Content tangentially related to surrounding content (sidebar)
<footer>	Footer for the document or section (copyright, contact info)
<div>	Generic block-level container (no semantic meaning)
	Generic inline container (no semantic meaning)

Heading Elements

HTML provides 6 levels of headings. Use them hierarchically to structure content:

```
<h1>Main Heading (largest)</h1>
<h2>Sub Heading</h2>
<h3>Sub-sub Heading</h3> ...up to <h6>
```

Paragraph & Text Elements

Tag	Purpose	Example
<p>	Paragraph	This is a paragraph.
 	Line break (empty tag)	Line 1 Line 2
<hr>	Thematic break (horizontal rule)	<hr>
	Bold / Important text	Bold
	Italic / Emphasized text	Italic
<small>	Smaller text	Fine print
<blockquote>	Long quotation (block)	<blockquote>Quote</blockquote>

2.2 Adding Images to a Website (Ch. 3 – HTML 83–94)

Image File Formats

Format	Pros	Cons	Best For
GIF	Small file; transparency; animation support	Limited to 256 colors	Icons, line drawings, animations
PNG	Lossless; transparency; 16M+ colors (PNG24)	No animation; larger than JPG for photos	Non-photo web images
JPG/JPEG	16.7M colors; lossy compression for photos	Larger file; loses quality on re-save	Digital photos
SVG	Scalable; code-based; no quality loss at any size	Not all features supported in older browsers	Shapes, icons, logos, animations

The Element

The element is an empty (void) element used to embed images. It requires two mandatory attributes:

```

```

Attribute	Required?	Description
src	Yes	Path or URL to the image file
alt	Yes	Alternative text for accessibility and if image fails to load
width	No	Width of the image in pixels
height	No	Height of the image in pixels
title	No	Tooltip text shown on hover

Image Paths

- **Absolute path: Full URL** – <https://www.example.com/images/photo.jpg>
- Relative path (same folder): photo.jpg
- Relative path (subfolder): images/photo.jpg
- Relative path (parent folder): ../photo.jpg

2.3 Exploring Div Elements (Ch. 3 – HTML 98–122)

A <div> (division) element is a generic block-level container used to group HTML elements and apply CSS styles to sections of a page. Divs have no visual appearance on their own.

```
<div id="header">  
  <h1>Welcome to My Site</h1>  
  <p>Subtitle text here</p>  
</div>
```

id attribute: Uniquely identifies ONE element on a page. Used with # in CSS. E.g., id="navigation"

class attribute: Applies to multiple elements. Used with . in CSS. E.g., class="highlight"

2.4 Adding Links to a Webpage (Ch. 3)

Links (hyperlinks) are created using the `<a>` (anchor) element. The `href` attribute specifies the destination.

Types of Links

```
<!-- External link (another website) -->
<a href="https://www.google.com">Visit Google</a>
```

```
<!-- Internal link (another page on same site) -->
<a href="about.html">About Us</a>
```

```
<!-- Anchor link (same page, different section) -->
<a href="#section2">Jump to Section 2</a>
<h2 id="section2">Section 2</h2>
```

```
<!-- Email link -->
<a href="mailto:info@example.com">Email Us</a>
```

```
<!-- Open in new tab -->
<a href="https://example.com" target="_blank">Open New Tab</a>
```

Attribute	Description
<code>href</code>	Destination URL or path (required)
<code>target="_blank"</code>	Opens link in a new browser tab/window
<code>target="_self"</code>	Opens in same tab (default)
<code>title</code>	Tooltip text on hover

2.5 Adding Lists (Ch. 3)

Unordered Lists (Bullet Lists)

```
<ul>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ul>
```

Ordered Lists (Numbered Lists)

```
<ol>
  <li>Step one</li>
  <li>Step two</li>
  <li>Step three</li>
</ol>
```

Description Lists

```
<dl>
  <dt>HTML</dt>
  <dd>Hypertext Markup Language - used to structure web content</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets - used to style web content</dd>
</dl>
```

Nested Lists

Lists can be nested inside one another. A nested list is placed inside an `<li>` element of the parent list.

```
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Banana</li>
    </ul>
  </li>
</ul>
```

2.6 Discovering Tables & Table Style (Ch. 8 – HTML 402–438)

HTML tables are used to display tabular data (data organized in rows and columns). Do NOT use tables for page layout – use CSS instead.

Basic Table Structure

```
<table>
  <caption>Monthly Sales Data</caption>
  <thead>
    <tr>
      <th>Month</th>
      <th>Sales</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>January</td>
      <td>$5,000</td>
    </tr>
    <tr>
      <td>February</td>
      <td>$7,500</td>
    </tr>
  </tbody>
</table>
```



```

</tbody>
<tfoot>
  <tr>
    <td>Total</td>
    <td>$12,500</td>
  </tr>
</tfoot>
</table>

```

Element	Description
<table>	Defines the table
<caption>	Table title/caption (placed right after <table>)
<thead>	Groups header row(s)
<tbody>	Groups body/data rows
<tfoot>	Groups footer row(s)
<tr>	Table row
<th>	Table header cell (bold, centered by default)
<td>	Table data cell

Table Spanning

Cells can span multiple columns or rows using attributes:

```

<!-- Span 2 columns -->
<td colspan="2">This cell spans 2 columns</td>

```

```

<!-- Span 3 rows -->
<td rowspan="3">This cell spans 3 rows</td>

```

Styling Tables with CSS

```

table { width: 100%; border-collapse: collapse; }
th, td { border: 1px solid #ccc; padding: 8px; text-align: left; }
th { background-color: #1a5276; color: white; }
tr:nth-child(even) { background-color: #f2f2f2; }

```

2.7 Creating Webpage Forms (Ch. 8 – HTML 423–438)

HTML forms collect user input and submit it to a server for processing. Forms are used for login pages, search boxes, registration forms, surveys, etc.

Form Structure

```

<form action="process.php" method="post">
  <!-- form elements go here -->
</form>

```

Attribute	Values	Description
action	URL	Where form data is sent (server-side script)
method	get / post	get: appends data to URL; post: sends data in request body (more secure)
enctype	multipart/form-data	Required when form includes file uploads

Common Form Elements

```
<!-- Text input -->
<label for="username">Username:</label>
<input type="text" id="username" name="username" placeholder="Enter username">
```

```
<!-- Password input -->
<input type="password" id="pwd" name="pwd">
```

```
<!-- Email input -->
<input type="email" id="email" name="email">
```

```
<!-- Radio buttons -->
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female
```

```
<!-- Checkboxes -->
<input type="checkbox" name="hobby" value="sports"> Sports
<input type="checkbox" name="hobby" value="music"> Music
```

```
<!-- Dropdown / Select menu -->
<select name="city">
  <option value="delhi">Delhi</option>
  <option value="mumbai">Mumbai</option>
</select>
```

```
<!-- Textarea -->
<textarea name="message" rows="5" cols="40"></textarea>
```

```
<!-- Submit button -->
<input type="submit" value="Submit">
<!-- OR -->
<button type="submit">Submit</button>
```

Input Type	Description
text	Single-line text field

password	Text field that hides characters
email	Email address field (validates format)
number	Numeric input
tel	Telephone number
url	URL input
date	Date picker
radio	Select one option from a group
checkbox	Select one or more options
file	File upload button
submit	Submit the form
reset	Reset form to defaults
hidden	Hidden data (not visible to user)

Form Accessibility

Always use <label> elements and associate them with inputs using the "for" attribute matching the input's "id". This improves accessibility for screen readers.

```
<label for="name">Full Name:</label>
<input type="text" id="name" name="name" required>
```

UNIT 3: CASCADING STYLE SHEETS (CSS)

Introduction • Inline, Embedded & External CSS • Selectors • Properties • Box Model • Layout

3.1 Introduction to CSS

Cascading Style Sheets (CSS) is a language used to describe the formatting/presentation of a document written in HTML. CSS separates content (HTML) from presentation (CSS), making websites easier to maintain and redesign.

CSS History

- CSS 1 (1996): Text, color, alignment, borders, padding, margins
- CSS 2 (1998): Positioning of elements
- CSS3 (2001+): Shadows, rounded borders, enhanced text effects, animations, media queries

3.2 Types of Style Sheets

CSS supports three ways to apply styles. The "cascade" refers to the order of priority – styles closer to the content take precedence.

1. Inline Styles (Highest Priority)

Applied directly to a single HTML element using the style attribute. Overrides all other styles.

```
<h1 style="color: navy; font-size: 24px;">Welcome</h1>
```

Use: When you need to format only ONE specific element in a unique way. Use sparingly – defeats the purpose of separating style from content.

2. Embedded (Internal) Style Sheets

Written inside <style> tags in the <head> section of an HTML document. Applies to that page only.

```
<head>
  <style>
    body { background-color: #f0f0f0; }
    h1 { color: blue; }
  </style>
</head>
```

Use: When styling a single page differently from the rest of the website.

3. External Style Sheets (Recommended)

A separate .css file linked to multiple HTML pages. Changing this one file updates ALL linked pages (reskinning).

```
/* styles.css */
body { font-family: Arial, sans-serif; margin: 0; }
h1 { color: #1a5276; }
```

Link in HTML <head>:

```
<link rel="stylesheet" href="styles.css">
```

Style Sheet Precedence (Cascade Order)

Priority from highest to lowest:

- **1. Inline styles (highest priority)**
- 2. Embedded (internal) style sheets
- 3. External style sheets
- 4. Browser default styles (lowest priority)

Inheritance: Child elements inherit styles from parent elements (e.g., paragraphs inherit body font).

Specificity: More specific selectors override less specific ones (e.g., p selector beats body selector).

3.3 CSS Syntax – Selectors & Declarations

Every CSS rule consists of a SELECTOR and a DECLARATION BLOCK.

```
selector {
  property: value;
  property: value;
}
```

Component	Description	Example
Selector	Identifies WHAT to style	body, h1, p, #nav, .highlight
Property	The style characteristic to change	color, font-size, background-color
Value	The specific setting for the property	blue, 16px, #ffffff
Declaration	One property:value pair	color: blue;
Declaration Block	All declarations inside { }	{ color: blue; font-size: 16px; }

3.4 CSS Selectors

Selector Type	Syntax	Description	Example
Element	element	Selects all elements of that type	p { color: black; }
ID	#id	Selects ONE element with that id (unique)	#header { background: blue; }
Class	.class	Selects all elements with that class	.highlight { color: red; }
Universal	*	Selects ALL elements	* { margin: 0; }
Descendant	a b	Selects b inside a	nav a { color: white; }
Group	a, b	Applies same style to multiple selectors	h1, h2, h3 { font-family: Arial; }
Pseudo-class	:state	Selects based on element state	a:hover { color: red; }

Common Pseudo-classes

Pseudo-class	Description
a:link	Unvisited link
a:visited	Visited link
a:hover	When mouse is over the element
a:active	When link is being clicked
input:focus	When an input is selected/focused
:nth-child(even)	Every even child element (useful for table row striping)

3.5 CSS Text Properties

Property	Description	Example Values
font-family	Sets the font typeface	Arial, Verdana, "Times New Roman", serif
font-size	Sets the font size	16px, 1.2em, 120%, x-large
font-weight	Sets font boldness	bold, bolder, lighter, 400, 700
font-style	Sets font style	normal, italic, oblique

color	Sets text color	blue, #1a5276, rgb(26,82,118)
text-align	Aligns text	left, right, center, justify
text-decoration	Adds decoration to text	none, underline, line-through
text-transform	Changes text case	uppercase, lowercase, capitalize
line-height	Sets line spacing	1.5, 24px, 150%
letter-spacing	Space between letters	2px, 0.1em

3.6 CSS Colors

Colors in CSS can be specified in multiple formats:

```
color: red; /* Named color */
color: #ff0000; /* Hexadecimal (6 digits) */
color: #f00; /* Hex shorthand (3 digits) */
color: rgb(255, 0, 0); /* RGB values (0-255) */
color: rgba(255, 0, 0, 0.5); /* RGB + Alpha (transparency 0-1) */
color: hsl(0, 100%, 50%); /* Hue, Saturation, Lightness */
```

3.7 CSS Box Model

Every HTML element is a rectangular box made of 4 layers (from inside out):

Layer	CSS Property	Description
Content	(width, height)	The actual text/image content
Padding	padding	Space BETWEEN content and border (inside the border, part of element background)
Border	border	The visible boundary around the element
Margin	margin	Space OUTSIDE the border (between elements, transparent)

```
div {
  width: 300px;
  padding: 20px; /* all sides */
  padding: 10px 20px; /* top/bottom left/right */
  border: 2px solid black; /* width style color */
  margin: 15px auto; /* top/bottom auto centers horizontally */
  box-sizing: border-box; /* includes padding+border in width */
}
```

Shorthand Properties

```
padding: 10px; /* all 4 sides = 10px */
padding: 10px 20px; /* top/bottom=10, left/right=20 */
padding: 10px 20px 15px; /* top=10, left/right=20, bottom=15 */
```

```
padding: 10px 20px 15px 5px; /* top right bottom left (clockwise) */
```

3.8 Inline Elements vs Block Elements

Block Elements	Inline Elements
Take up full width available	Only take up necessary width
Start on a new line	Stay on the same line as content around them
Can have width/height set	Width/height ignored in CSS
Examples: <div>, <p>, <h1>–<h6>, , , <table>	Examples: , <a>, , ,

3.9 Creating & Linking an External CSS File

Create a CSS File

Create a text file with the .css extension (e.g., styles.css). No HTML tags inside!

```
/* styles.css - Comments use this format */

/* Reset */
* { box-sizing: border-box; margin: 0; padding: 0; }

/* Body styles */
body {
  font-family: Arial, sans-serif;
  font-size: 16px;
  color: #333333;
  background-color: #ffffff;
}
```

Link CSS to HTML

Add this inside the <head> element of every HTML file:

```
<head>
  <link rel="stylesheet" href="styles.css">
</head>
```

Attribute	Value	Meaning
rel	stylesheet	Specifies the relationship (this is a style sheet)
href	styles.css	Path to the CSS file

3.10 CSS Webpage Layout

Float and Clear Properties

The float property was traditionally used to create column layouts by floating elements left or right.

```
.sidebar { float: left; width: 30%; }
.main-content { float: left; width: 70%; }
.footer { clear: both; } /* clear: stops floating above */
```

ID and Class Selectors for Layout

Use id for unique page sections and class for reusable styles:

```
/* CSS */
#navigation { background-color: #333; }
#content { margin: 0 auto; max-width: 1000px; }
.highlighted { background-color: yellow; }
.btn { padding: 10px 20px; border-radius: 5px; }
```

Structural/Semantic Element Selectors

You can target HTML5 semantic elements directly as selectors:

```
header { background-color: #1a5276; color: white; padding: 20px; }
nav { background-color: #2e86c1; }
nav a { color: white; text-decoration: none; padding: 10px; }
main { max-width: 900px; margin: 0 auto; padding: 20px; }
footer { background-color: #333; color: #ccc; text-align: center; }
```

Common Background Properties

Property	Description	Example
background-color	Sets background color	background-color: #f0f0f0;
background-image	Sets background image	background-image: url("bg.jpg");
background-repeat	Controls if image repeats	background-repeat: no-repeat;
background-position	Positions background image	background-position: center top;
background-size	Controls image size	background-size: cover;

3.11 CSS Comments & Validation

Comments in CSS explain the code. Browsers ignore comments.

```
/* This is a CSS comment */
/* Use comments to organize sections:
   Header styles
   Navigation styles */
```

Validating CSS

The W3C provides a free CSS Validator at: <https://jigsaw.w3.org/css-validator/>

Validation checks for: syntax errors, invalid property names, invalid values, and browser compatibility issues. Always validate before publishing.

QUICK REFERENCE — HTML TAGS & CSS PROPERTIES

Essential HTML Tags Summary

Category	Tags
Document Structure	<!DOCTYPE html>, <html>, <head>, <meta>, <title>, <body>
Semantic Layout	<header>, <nav>, <main>, <section>, <article>, <aside>, <footer>, <div>
Headings	<h1>, <h2>, <h3>, <h4>, <h5>, <h6>
Text	<p>, , <hr>, , , , <small>, <blockquote>
Links	
Images	
Lists	, , , <dl>, <dt>, <dd>
Tables	<table>, <caption>, <thead>, <tbody>, <tfoot>, <tr>, <th>, <td>
Forms	<form>, <input>, <label>, <textarea>, <select>, <option>, <button>

Essential CSS Properties Summary

Category	Properties
Text & Font	font-family, font-size, font-weight, font-style, color, text-align, text-decoration, line-height
Background	background-color, background-image, background-repeat, background-size, background-position
Box Model	width, height, padding, margin, border, border-radius, box-sizing
Layout	display, float, clear, position, top, left, right, bottom, z-index
Flexbox	display: flex, flex-direction, justify-content, align-items, flex-wrap
Color	Named, #hex, rgb(), rgba(), hsl()
Selectors	element, #id, .class, *, a b, a > b, a:hover, a:nth-child()

— End of Notes —